

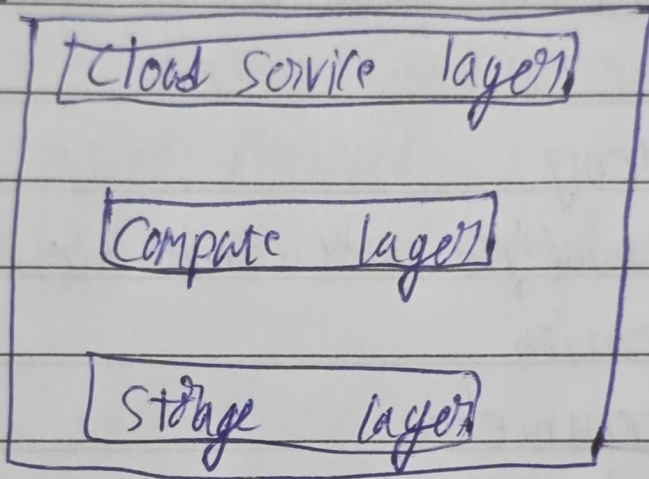
Snowflake (by learning thing (youtube))

DATE:

Q) What is snowflake?

- It is a cloud based data platform to store, process & analyze large amount of data
- It completely runs on the cloud providers

Architecture:-



Key features:-

- stores data in Columnar
- micro partition → Data will be split small chunks (~10MB)
Automatically managed.
- Time travel
- Independent Compute
- Auto Scaling
- Caching
- metadata management

Why this architecture is powerful :-

1. Separation of Storage & Compute.

* Normally if we want to increase storage means we have to increase Compute also it will costly

* Now this architecture allows us to increase independently.

2. High Concurrency

* At a time multiple users can use.

3. Cost optimization

4. Zero maintenance.

Q) What is virtual machine?

Ans) It is a compute engine it runs our VMs.

Type of Clusters :-

1. Single cluster

2. Multiple cluster.

Snowflake Data loading types:-

1. Snowflake UI

2. Bulk loading (copy INTO)

3. Continuous loading (snowpipe)

- Snowpipe automatically loads files when they arrive in cloud storage.

4. Real-Time streaming (Snowpipe streaming)

- No file staging required

- Data directly streams into snowflake

File format:

* How should I read this file?

* Whenever Snowflake loads or unloads files, it uses a file format object.

Ex:- Create File format file-format-name

TYPE = CSV

FIELD_DELIMITER = ','

SKIP_HEADER = 1

DATE_FORMAT = 'DD-MM-YYYY';

Types of Tables

1. Permanent table:-

- * Default table type
- * Used for critical business data that needs long-term storage & strong recovery.
- * stores in internal snowflake storage
- * Time travel ✓, fail safe ✓
- * Default retention period 1 day (we can go upto 90 days)
- * Highest cost

ex: create table table-name ();

- * Production, Gold layer, Fact & dimension layer, reporting & analysis

2. Temporary table:-

- * exists only for the duration of the session, Automatically dropped when session ends.
- * stores in internal snowflake storage.
- * Time travel (limited during session), X fail safe.
- * Lowest cost

ex: Create temp table table-name ();

- * Temporary processing, testing, experimentation

3. Transient table:-

- * Similar to permanent table without fail safe.
- * Used for staging, intermediate & re-creatable data.
- * Store in internal snowflake storage.
- * Time travel ✓, fail safe X
- * Default retention period 1 day (it can go upto 90 days)
- * Lower Cost
- * Ex: create transient table table_name ();
- * Staging, bronze layer, ETL intermediate data.

4. External table:-

- * Data remains in your cloud storage. Snowflake only stores meta data & file pointers.
- * Data stores in external storage.
- * Time travel X, fail safe X
- * ~~High~~^{very low} Cost
- * Ex: create external table table_name ()
Location = ' ' ;

5. Dynamic table:

- * Automatically refreshed table based on a query.

- * Snowflake manages incremental updates.

- * Data stores in internal snowflake storage

- * Time travel ✓, fail safe ✓

- * Retention period (1-90 days)

- * Higher cost

- * ex:- create dynamic table table_name

target_Lat = '5 min'

AS

select * from - - -

- * Incremental ETL, Silver, Gold, Near real-time pipeline,

Reduce manual refresh jobs.

6. ICEBERG Table:

- * Uses apache Iceberg open table format

- * Stored in your data lake

- * Data stores in internal snowflake storage.

- * Time travel & fail safe depends on Iceberg.

- * Retention period depends on Iceberg.

ex:- create iceberg table table_name () catalog = catalog_name
external-volume = 'volume'.

- * Low to medium cost.

Stage in snowflake

A stage is a file storage location used by snowflake to temporarily store or access files before loading them into tables or unloading data from tables.

- * Landing zone for files before snowflake pools data.

Types:-

1. Internal stage:-

- * Files are stored inside snowflake-managed storage.
- * snowflake manages everything, you don't manage bucket directly.

(A) user stage:- (@~)

- * Each user automatically gets one. No need to create.

(B) Table stage:- (@%^{stage}~~table~~-name)

- * Each table automatically gets one. No need to create.
- ex:- ~~copy~~ Copy into table-name from @%-stage-name

(C) Named Internal stage:-

- * It is a scalable snowflake-managed stage created manually.

ex:- create stage stage-name;
use case:- multiple files, batch ingestion, scalable pipeline

2. External Stage:

* File already in cloud storage in external

Ex: Create stage stage-name

URL = ' ' ,

storage-integration = storage-integration-name;

SNOWSQL

* It is a CLI

* It let's you connect to Snowflake from your local machine or server & run commands

* You type SQL in terminal insted of UI.

Use Cases:-

* automation, local file upload, scripting, scheduling jobs,

CI/CD pipelines & server based execution.

Copy into Command

* It is a bulk data loading & unloading Command.

* Loads data from files/stage into tables

* Whenever file arrives into stage (internal stage, table stage, extn stage) snowflake also copy into command to move data from the file into a table.

Ex:-

Copy into table-name

from @stage

file-format = () ;

* It supports the incremental load.

* It automatically identify new file.

To process bad records:-

Copy into table-name

from @stage

file-format = ()

on-error = 'continue';

Steps to create external stage:-

1. Create table

2. Create S3 bucket

3. Create file format

4. Create External stage using AWS Access Key.

Ex:- Create or replace stage stage-name

URL = ' ';

Credentials = (

AWS_KEY_ID = ' ';

AWS_SECRET_KEY = ' ';

file-format = () ;

DATE:

Storage integration:-

create or replace storage integration storage integration name

type = 'external-stage'

storage-provider = 'S3'

enabled = TRUE

storage-aws-role-arn = ' ';

storage-allowed-location = (' ');

Snowpipe

It is Snowflake's continuous data ingestion service that automatically loads new files into Snowflake as soon as they arrive in a stage.

Advantages:-

- * Automatic loading

- * No scheduler required

- * Near real-time data

- * Data will available within seconds/min

- * Serverless $\rightarrow$ No warehouse needed for ingestion

- * Incremental loading $\rightarrow$ Tracks loaded files automatically.

- * Scalable $\rightarrow$ Handling thousands of files.

Ex:- Create or replace pipe pipe-name

AUTO-INTEST = True

AS

Copy into table-name from @stage;

External table

An external table allows snowflake to query data stored in external storage

* Data remains in S3, snowflake uses the metadata

Ex:- Create or replace external table table-name

Location = ^{external-}@stage-name

file-format = ();

Pros:-

1. No Data load required
2. Lower ^{storage} cost
3. Good for data lake
4. Easy access to new files.

Cons:-

1. Slower performance
2. No DML operation.
3. metadata refresh needed

Ex:- Alter External table table-name refresh;

Snowflake caching:-

Snowflake stores previously used data/segment so that future queries run faster & cost less.

Types:-

1. Result Cache:-

* Must be:- Same query, same underlying data, same permissions, within 24 hours.

Benefits:-
* Almost instant response, no warehouse computation, no credits consumed.

2. Warehouse Cache:-

* Data is downloaded from snowflake storage to warehouse SSD cache.

* Snowflake may read data from warehouse SSD cache.

Ex:- select * from sales;

select sum(amount) from sales;

* Cache is cleared when warehouse suspended.

* Benefits:- Faster queries, less data scanning.

3. metadata cache:-VIEWSTypes:-1. Standard view:-

- * Stores only SQL definition
- * Every query executes the underlying SQL

ex:- Create view view-name as

select - - - -

2. Secure view:-

- * Used for data security
- * Normal view may expose underlying query details.

hides:-

- * Underlying table structure
- * Query logic
- * Sensitive business data.

ex:- Create secure view view-name as

select - - -

3. Materialized view:-

- * It stores query & data, ^{In background snowflake} * automatically do refresh.

Advantages:-

- * Fast reporting
- * Reduced computation
- * Useful for dashboard

Disadvantages:-

- * Consumes storage.

ex:- Create materialized view view-name as
select - - -

Zero-Copy Clone

- * It is a snowflake feature that allows you to create a copy of a table, schema or database without physically copying the data.
 - * The clone is created almost instantly because snowflake only copies metadata & references existing micro partitions.
 - * It is like shallow clone.
- ex:- Create table table-name clone ~~target~~ source-table-name;
- ↓
We can create schema / Database
- Limitation -> We can't see "Internal Stage & External Table"

Snowflake streams

It is a snowflake object that track changes made to a table

* It tells what was inserted/updated/deleted?
without scanning the entire table.

Ex:- Create a replace stream stream-name on table table-name;

For update:-

* Records will appear 2 time 1 for del 1 for up

1. metadata\$action → Insert, metadata\$update → True
2. metadata\$action → delete, metadata\$update → True

* 2 one is for reference, while performing SQL 2 we consider 1 only.

For Insert:-

metadata\$action → Insert, metadata\$update → False

For delete:-

metadata\$action → Delete, metadata\$update → False.

* one more type stream also there append-only.

* It only tracks insert data.

Ex:- Create stream stream-name on table table-name
APPEND-ONLY = True;